

Profile

Nuclear Engineering M.S. candidate and Graduate Research Associate with a 4.0 GPA and multidisciplinary training in physics, software engineering, and computational nuclear materials. Develops Python-based rate-theory and stiff-ODE models for irradiation-induced defect evolution, diffusion kinetics, and dislocation-loop behavior. Experienced in nonlinear parameter estimation, experimental-data fitting, model validation, and reproducible scientific computing.

Technical Skills

Scientific Computing	Python, NumPy, SciPy, Pandas, Matplotlib, Jupyter, MATLAB
Numerical Methods	Stiff ODE systems, nonlinear least squares, multistart optimization, parameter estimation, regression, model validation
Nuclear & Materials	Radiation damage, rate theory, defect kinetics, dislocation loops, diffusion processes, reactor physics, fuel-cycle analysis
Software & Data	Git, C, Java, SQL, $\LaTeX$, BigQuery ML, scientific data processing and visualization

Research Experience

Graduate Research Associate	2025–Present
The Ohio State University, TME Group	Columbus, Ohio
- Develop physics-based rate-theory models for irradiation-induced point-defect evolution and dislocation-loop dynamics in uranium-based nuclear materials. - Implement coupled stiff ODE systems in Python to simulate defect production, recombination, diffusion, loop nucleation, growth, coalescence, and sink interactions. - Fit multi-temperature irradiation datasets using nonlinear least-squares and multistart optimization to estimate diffusion coefficients, activation energies, and kinetic parameters. - Compare competing mechanisms against transmission electron microscopy measurements of loop radius and density to identify physically consistent model scenarios. - Build reproducible analysis workflows and publication-quality figures using Python, SciPy, Git, Jupyter, and $\LaTeX$.	
Undergraduate Researcher - Thorium Fuel-Cycle Modeling	May–Dec. 2024
Universidad de los Andes	Bogota, Colombia
- Developed computational models of breeding performance, transmutation, neutronic behavior, and isotopic evolution in thorium-based fuel cycles. - Compared thorium and conventional uranium fuel cycles and evaluated implications for advanced reactor concepts.	
Computational Nuclear Astrophysics Researcher	Nov. 2023–June 2024
Universidad de los Andes	Bogota, Colombia
Modified nuclear-reaction-network code to improve numerical stability and asymptotic abundance calculations; evaluated sensitivity to changing physical conditions.	

Education

M.S. in Nuclear Engineering	Expected May 2027
The Ohio State University	GPA: 4.0/4.0
B.S. in Systems and Computing Engineering	July 2026
Universidad de los Andes	Bogota, Colombia
B.S. in Physics	April 2025
Universidad de los Andes	Bogota, Colombia

Publications

Co-author. “Impact of Radiation Damage in Uranium Dioxide: Implementing Rate Theory for Defect Evolution and Thermal Conductivity in Post-Irradiated Annealed Uranium Dioxide.” Manuscript submitted to the *Journal of Nuclear Materials*.

Additional Experience & Credentials

Teaching	Physics Teaching Assistant, Universidad de los Andes, January–June 2024
Training	Google Cloud skill badges in BigQuery ML predictive modeling and machine-learning data preparation
Languages	Spanish: Native English: Advanced professional proficiency